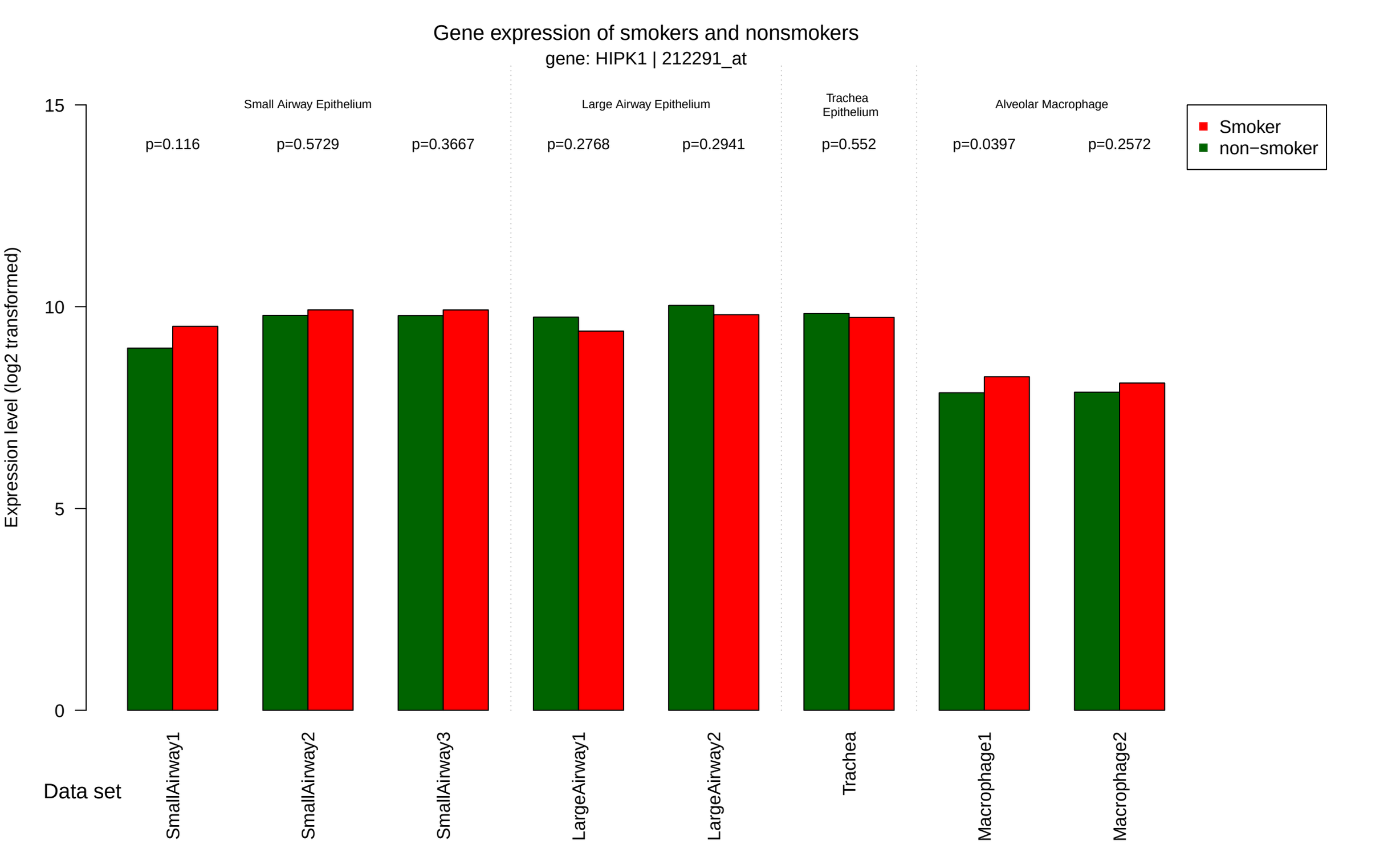
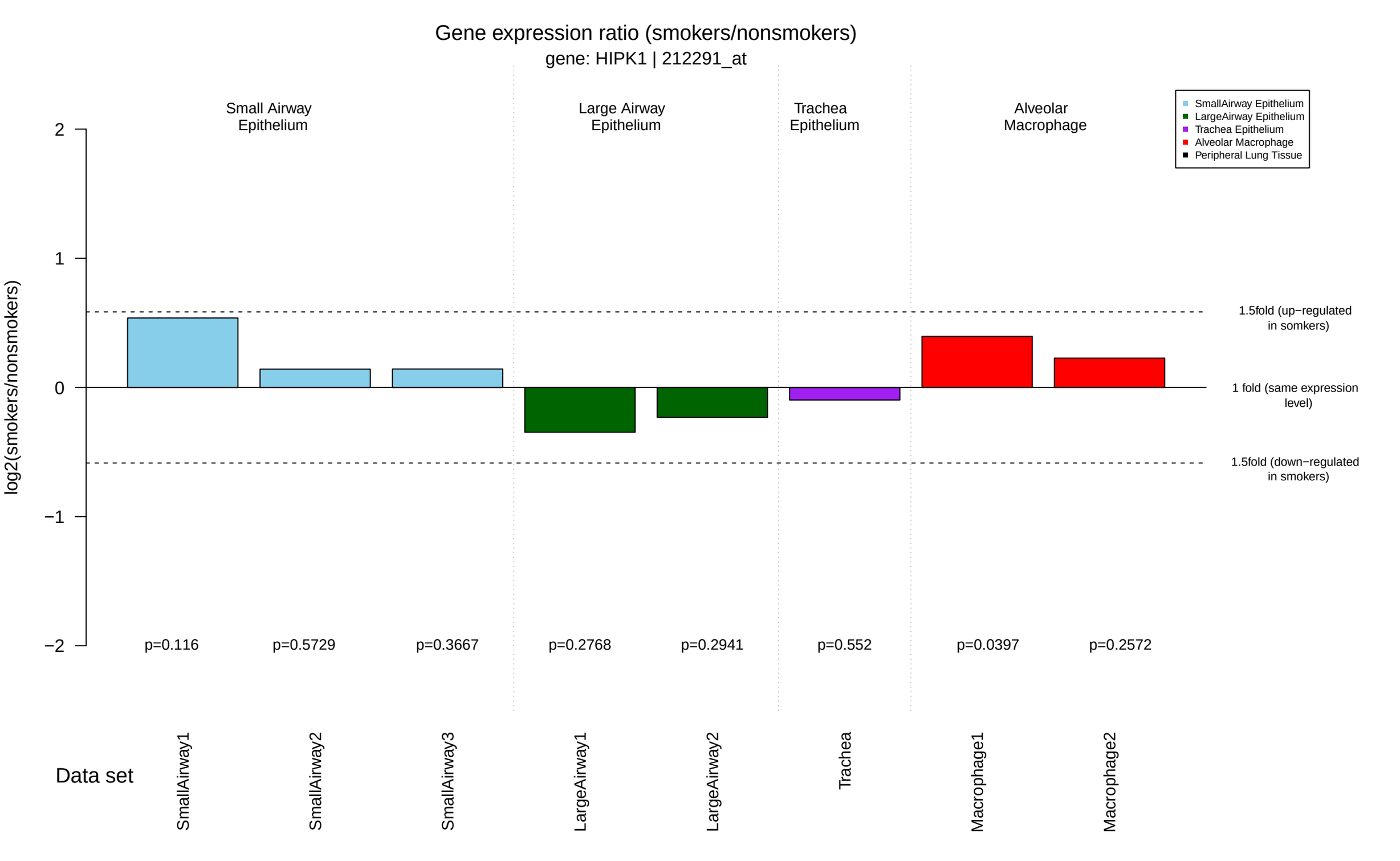
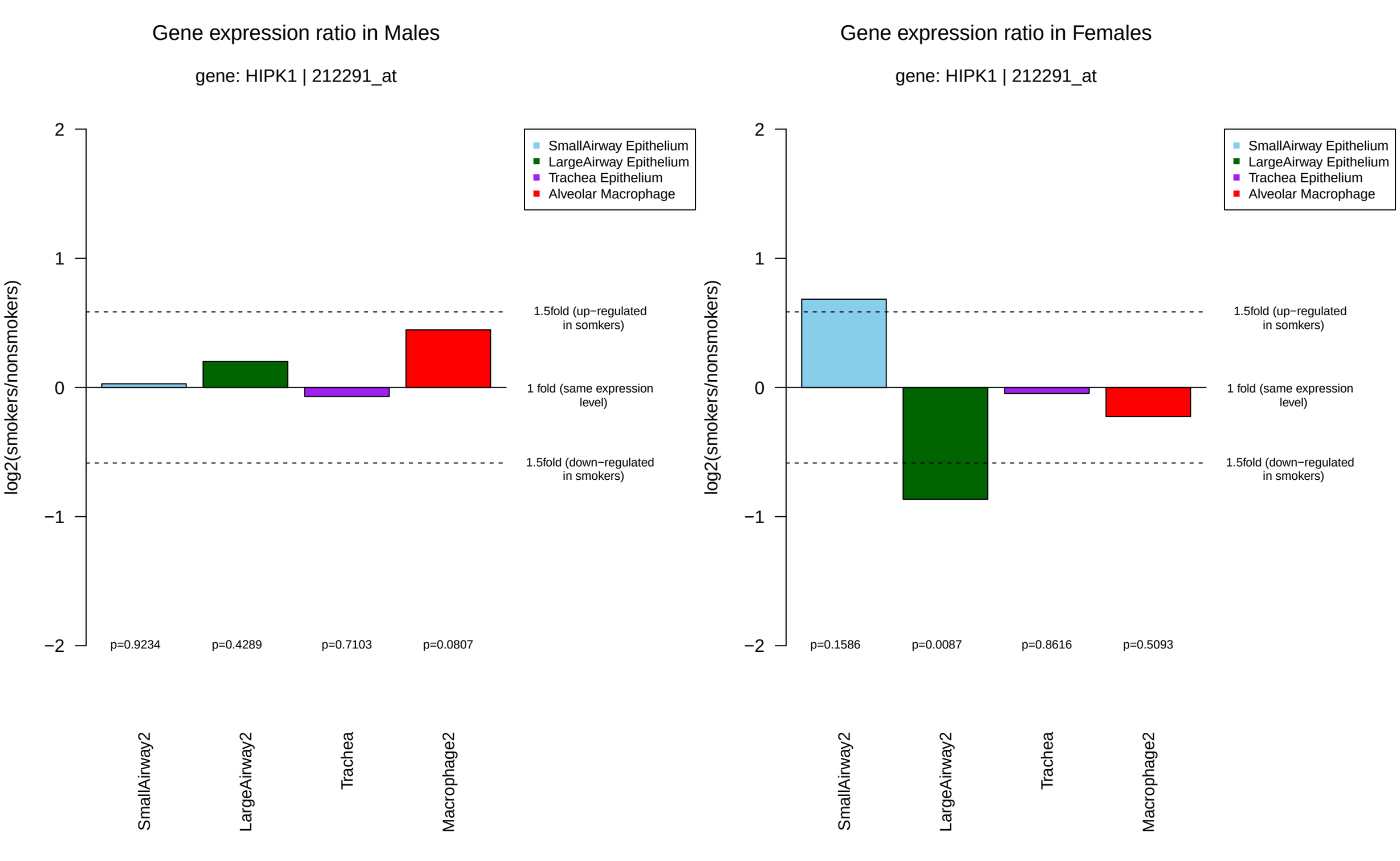
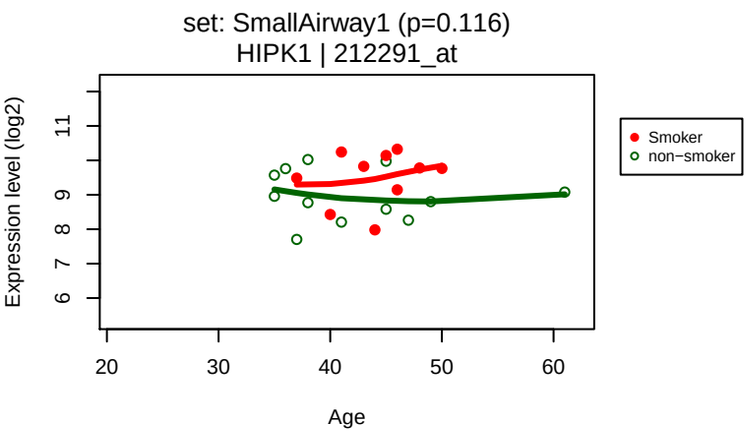
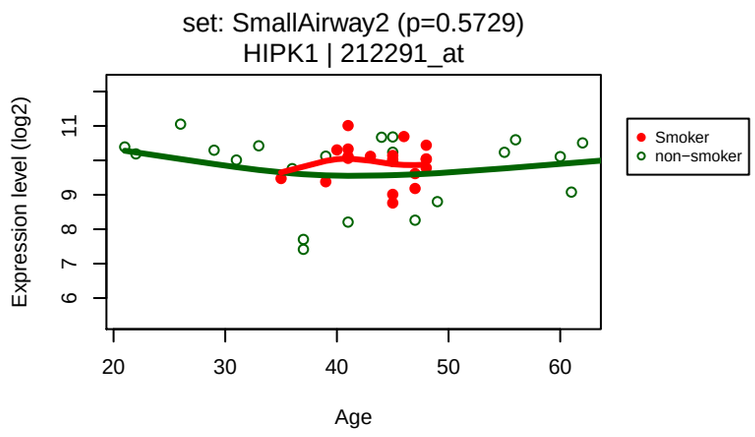
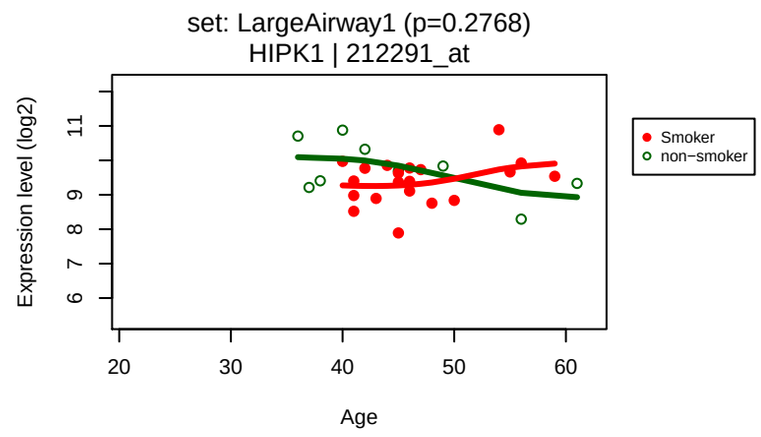

Gene expression of smokers and nonsmokers
gene: HIPK1 | 212291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.116
p=0.5729
p=0.3667
p=0.2768
p=0.2941
p=0.552
p=0.0397
p=0.2572
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

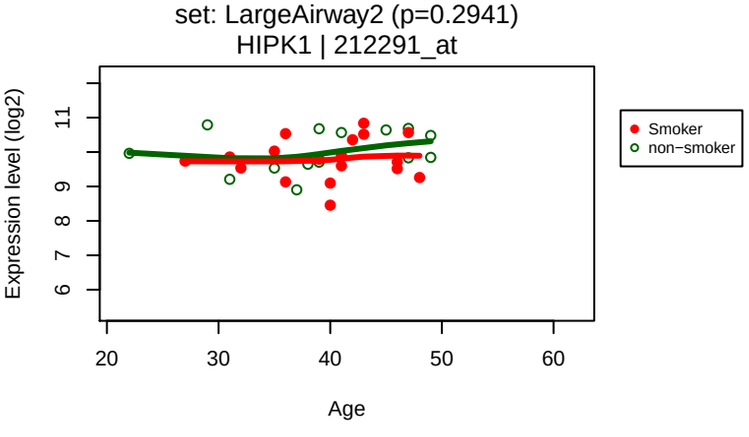

Gene expression ratio (smokers/nonsmokers)
gene: HIPK1 | 212291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.116
p=0.5729
p=0.3667
p=0.2768
p=0.2941
p=0.552
p=0.0397
p=0.2572
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

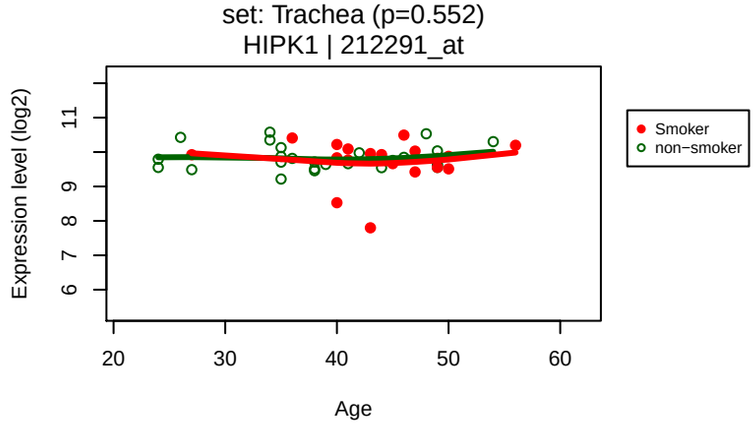

Gene expression ratio in Males
gene: HIPK1 | 212291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9234
p=0.4289
p=0.7103
p=0.0807
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK1 | 212291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1586
p=0.0087
p=0.8616
p=0.5093
SmallAirway2
LargeAirway2
Trachea
Macrophage2

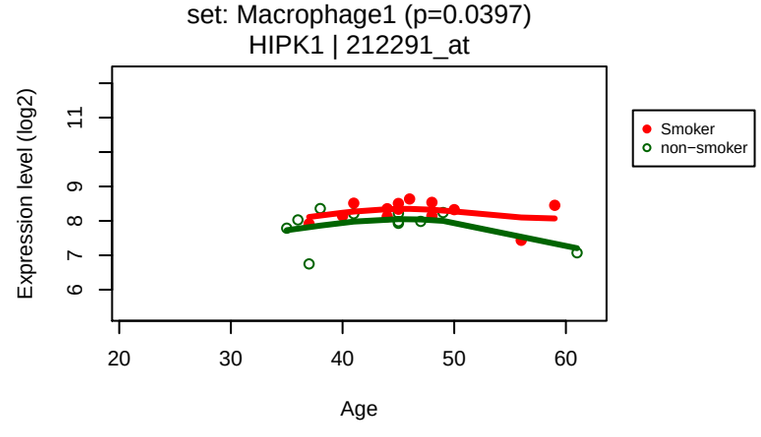

set: SmallAirway1 (p=0.116)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker

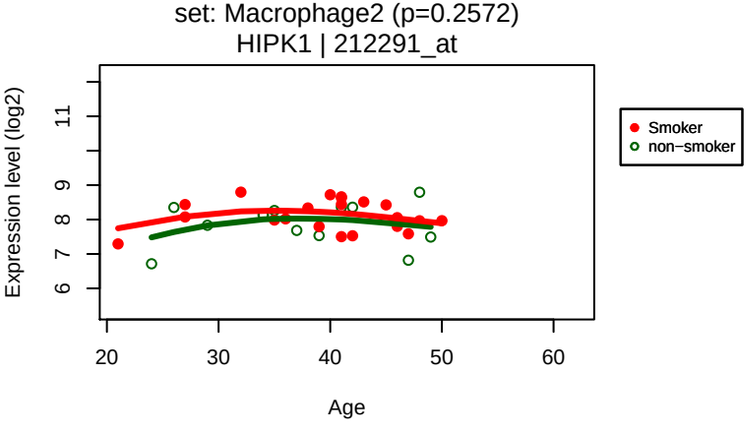

set: SmallAirway2 (p=0.5729)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker

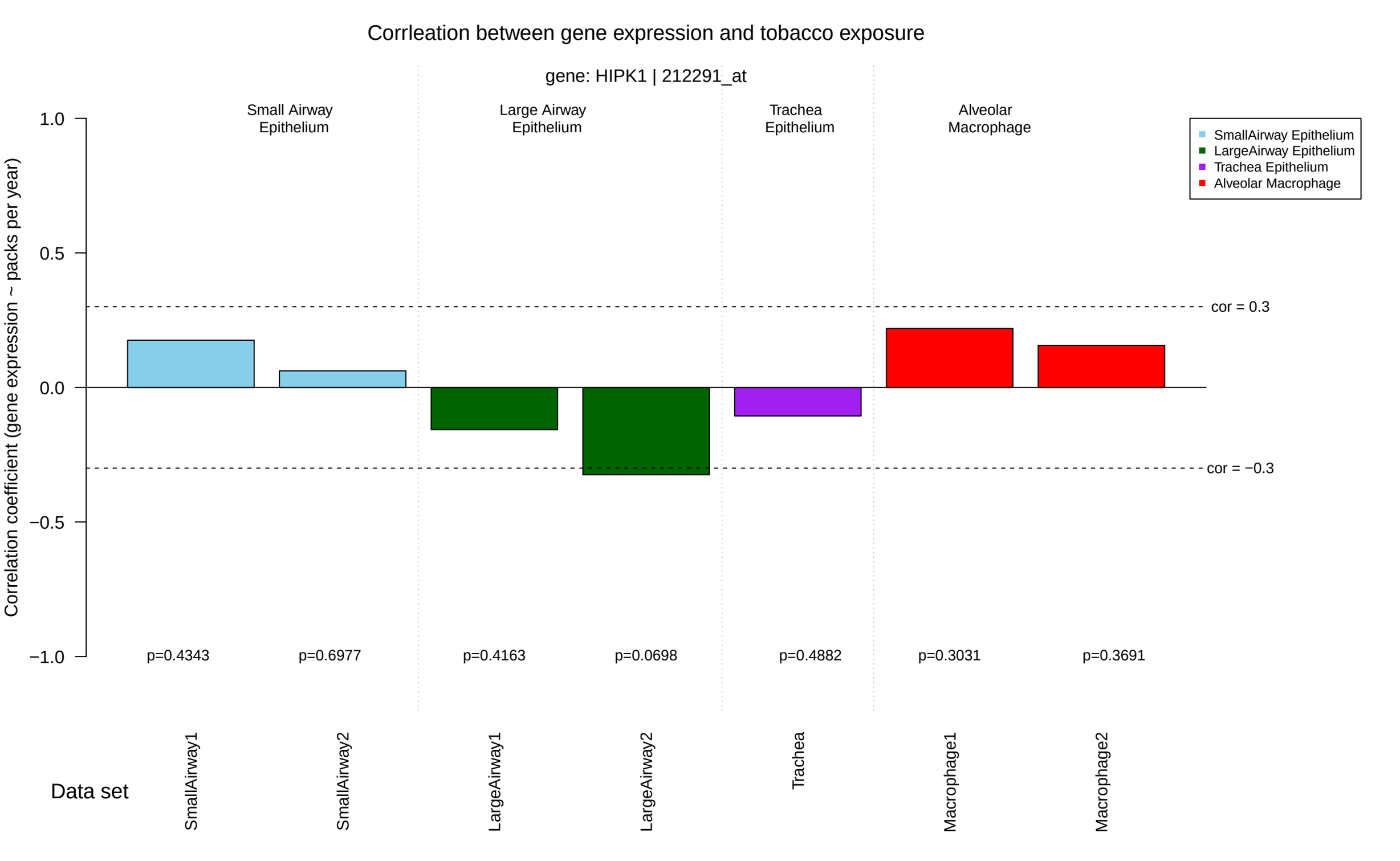

set: LargeAirway1 (p=0.2768)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2941)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.552)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0397)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2572)
HIPK1 | 212291_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK1 | 212291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4343
p=0.6977
p=0.4163
p=0.0698
p=0.4882
p=0.3031
p=0.3691
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2